

Assignment 3 - RAID Structure

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1. Introduction

Persistent storage is a vital component of modern computing systems, and with the increasing amount of data being stored, the demand for solutions that meet speed, reliability, and cost requirements is growing. No single hard drive can be both high performing and rugged, and when data loss or downtime is not an option, one drive just isn't sufficient.

The RAID means Redundant Array of Independent Disks. The concept is to merge several drives and arrange them in a manner that gives you increased performance, increased fault tolerance or both. Originally the letter I was used for "inexpensive" as a high-end, single drive alternative, but today it represents independent, as the drives cooperate in the array. RAID solves two issues: reliability – if one drive fails, RAID keeps data safe, and performance – if multiple drives are used, RAID can serve data faster. These are both accomplished using two basic methods: redundancy and striping..

2. Core Concepts: Redundancy and Striping

Consider failure rates at scale, and you'll understand the need for redundancy. An array of 100 disks with a mean time between failures (MTBF) of 100,000 hours statistically fails once every 1,000 hours, or 42 days. If not protected, each failure results in data loss. The answer is to keep redundant data around so that, if a drive fails, the lost data can be recovered from the remaining working drives.

The easiest way to do this is to have a mirror: keep a copy of each drive. All writes to both original and mirror. So if one's down, the other's up. A per-drive MTBF of 100,000 hours and a mean repair time of 10 hours puts the calculated MTTDL for a mirrored pair in the hundreds of millions of hours, compared to just one drive. A practical one is the torn write: if the machine lose its power in the middle of a write, the two copies are not the same. That's normally done with a nonvolatile write-back cache on the RAID controller that ensures the write is preserved during a power loss.

Striping spreads data over several drives to make these drives work in parallel. For block-level striping blocks of a file are distributed round-robin between N drives. Each of the multiple drives can then service multiple I/O requests at the same time, thereby improving throughput. The response time for large sequential reads and writes (all drives working together) and throughput for many small concurrent requests (the load is distributed across drives) is improved by striping. All modern RAID configurations use the standard form of block-level striping.

3. RAID Levels

Combining striping, mirroring, and a third technique called parity in different ways produces a set of configurations known as RAID levels. Parity uses XOR arithmetic on each corresponding data block

on both drives to generate a small redundancy block. If one drive fails, the data on the missing drive can be reconstructed from the remaining drives and the parity. The different RAID levels offer different compromises between storage efficiency, performance and fault tolerance.

RAID 0 — Pure Striping

RAID 0 is a RAID configuration that does not provide any redundancy. This provides the maximum read and write performance with no wasted storage space for parity or mirroring. The downside is that it's completely vulnerable: If one drive fails, all information on the array is lost, as each drive contains different parts of each file. RAID 0 is only suitable for applications that require fast access to data, such as video editing scratch space, simulation buffers or caches where data can be easily reconstructed.

RAID 1 — Mirroring

RAID 1 maintains a redundant copy of all drives. Both drives can be read from at the same time and recovery from a failure is simple: read in from the other drive and copy over. Rebuild is the quickest of all the RAID levels. The drawback is that half of the overall drive capacity is used up by the mirror. For critical systems like boot drives or small databases, recovery speed and cleanliness are paramount, RAID 1 is ideal.

RAID 4 — Block Striping with Dedicated Parity

RAID 4 distributes data among N drives and parity information to a single special drive. Reads are efficient as they typically access only a single data drive. Large reads and writes are fast, all data drives are in parallel. The downside is that all the writes, regardless of which data drive is being used, need to be written to the dedicated parity drive. This forms a drive parity problem. The cost of small writes is particularly high because they need to perform a read-modify-write sequence: Read old data block, read old parity, calculate new parity, write new data block, write new parity, 4 operations per drive access. Although RAID 4 is not widely used these days, it has one benefit: new drives can be added to the array without affecting the data, because the parity values will not change when new drives are added.

RAID 5 — Distributed Parity

RAID 5 overcomes the parity bottleneck of RAID 4 by spreading parity blocks around in a rotating fashion. Each stripe has a unique drive that is used to store its parity, which evenly distributes the load of writes. RAID 5 is the most popular parity RAID level. N+1 drives gives the same performance as N drives and will remain operational in case of failure of one drive. Read performance is very good and write performance is moderate. Only 1/N of the total capacity is used for storage. RAID 5 is ideal for small to medium sized databases, NAS devices, and file servers in general.

RAID 6 — Double Distributed Parity

RAID 6 is an extension of RAID 5 and has two independent redundancy blocks per stripe, calculated differently to give different recovery information. This allows the system to endure two drive failures, instead of one. This is important for large arrays when a second drive may fail in the middle of the rebuild of the first drive. There's a bit more overhead at 2/N (2N), and writes take longer because there are two parity computations. RAID 6 is the most popular RAID configuration used in today's enterprise storage arrays.

RAID 1+0 — Mirrored Stripes

RAID 1+0 (or RAID 10) is a combination of mirroring and striping. First mirrored in pairs, then striped in pairs. When a drive fails, only the mirrored drive is impacted and the remaining drives continue to operate normally. Multiple failures could be allowed to happen provided that they don't occur simultaneously on both drives in the same mirror pair. RAID 10 provides high performance and fault tolerance, however, with 50% storage overhead due to mirroring. It is the default one for applications that have a high transaction rate, such as OLTP databases and heavily loaded web servers.

4. Comparison of RAID Levels

The table below summarizes the key characteristics of the major RAID levels discussed above.

RAID Level	Min Drives	Fault Tolerance	Read Perf.	Write Perf.	Storage Efficiency
RAID 0	2	None	Very High	Very High	100%
RAID 1	2	1 drive failure	High	Moderate	50%
RAID 4	3	1 drive failure	High (reads)	Low (writes)	(N-1)/N
RAID 5	3	1 drive failure	High	Moderate	(N-1)/N
RAID 6	4	2 drive failures	High	Lower	(N-2)/N
RAID 10	4	Multiple (diff. pairs)	Very High	High	50%

5. Implementation and Additional Features

RAID can be set up at different layers of a computing system. Software RAID includes the RAID logic within the existing OS and CPU and hardware, providing flexibility and cost-efficiency, but requiring CPU cycles to calculate parity. Hardware RAID relies on a dedicated controller (HBA) that has its own processor, NVRAM cache and parity hardware, which takes load off of the main CPU.

The most common way in enterprise environments is to use storage arrays: a single array of drives, controller and cache that provides ready-to-connect volumes to any attached server, obscuring all RAID details from a clean front-end interface. A hot spare is a physically installed drive that is not currently in use in the array. When a live drive fails, the RAID controller will start rebuilding without waiting for a tech to arrive and take the drive out of the RAID array. This reduces the vulnerability window, the time when there is less redundancy in the array and it is more vulnerable, from days to hours. Several hot spares can be employed to handle more than one failure.

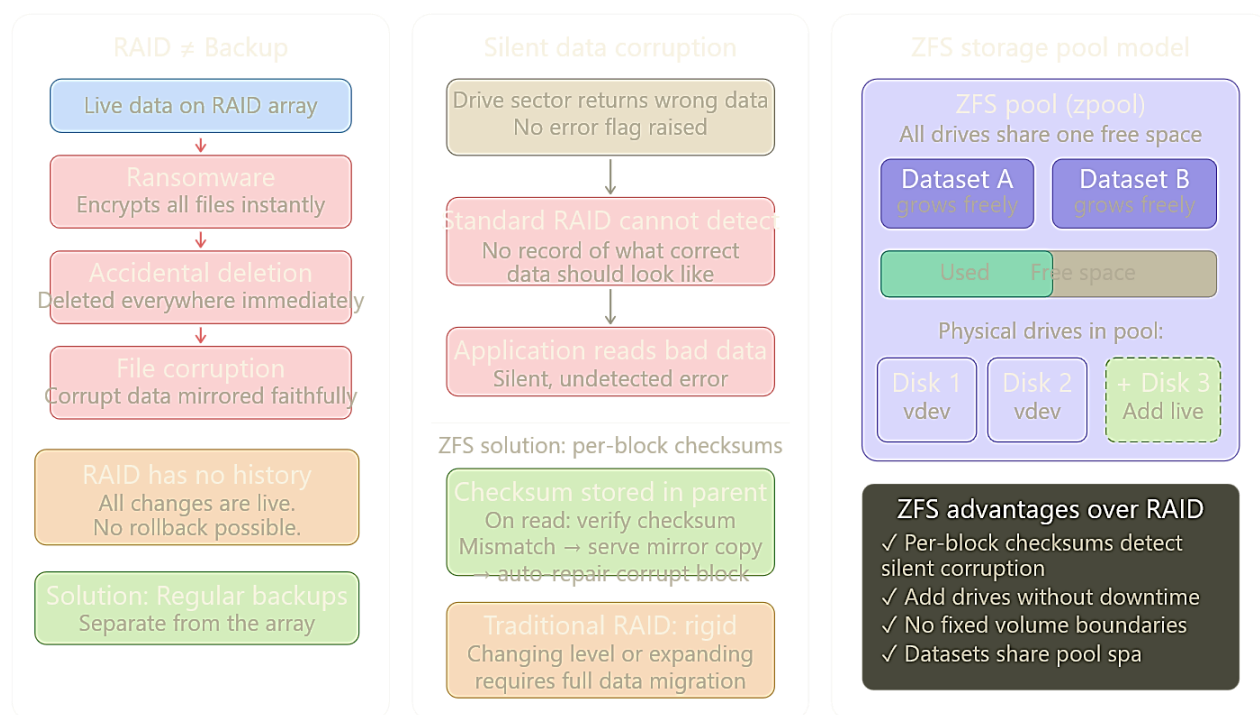
Replication extends redundancy to a second site by replicating writes to a second location. Synchronous replication is when a write is only confirmed when it has been complete at both sites (adds latency, but also means no data loss). Asynchronous replication gathers writes and then sends them at intervals, which speeds up the process and isn't dependent on distance, but means that the latest changes might be lost if the primary site goes down. The most important safeguard against site-level disasters that could affect even a fully redundant local array is replication.

6. Limitations of RAID

The first and most crucial point to grasp about RAID is that it's not a back-up. It won't protect from accidental deletion, file corruption, ransomware or software bugs. Changes to the live data are instantly mirrored throughout all mirrors and parity blocks. There's no way back, that's why backups are crucial even with a fully RAID-protected system.

Silent data corruption is a less obvious issue where individual sectors on a drive provide incorrect information and do not report a failure. This can't be detected by standard RAID, as it doesn't remember what the data should be. Modern file systems such as ZFS solve this issues by calculating a checksum for each block of data and each block of metadata and putting them in the block's parent pointer. A checksum is checked for a block when it is read. Unlike RAID, ZFS will also serve the correct data if corruption is detected, and fix it automatically, if it is mirrored.

Traditional RAID configurations are also not very flexible, after they have been configured, switching RAID levels or adding more drives may involve transferring all data off, reformatting, and transferring all data back on again. ZFS does this by integrating volume management and file system management into one layer. Drives are attached to a storage pool and the free space available on the drives attached to the storage pool is dynamically shared by all file systems. The volume boundaries are not fixed and as more drives are added, so can storage capacity grow without impacting the data.



7. Conclusion

RAIDs are a key element of contemporary storage design. RAID offers various configurations that can provide different levels of performance, redundancy, and cost-effective fault tolerance. RAID 0 provides no protection, but it does offer greatest throughput. RAID 1 and 10 are designed to focus on reliability and quick recovery. For most server and enterprise applications, RAID 5 and 6 are a sensible compromise.

The right level depends on understanding the application's requirements: how critical is the data, what is the required throughput, how much time is allowed for the system to be in a degraded state during rebuild and what budget is available. It is equally important to know what RAID cannot do. It's not a backup solution by itself, nor is it able to identify any silent data corruption. RAID along with regular backups and checksumming file systems, such as ZFS, offers the best level of data protection. It is essential knowledge in OS and storage system design to fully understand RAID, its features, and its limitations.